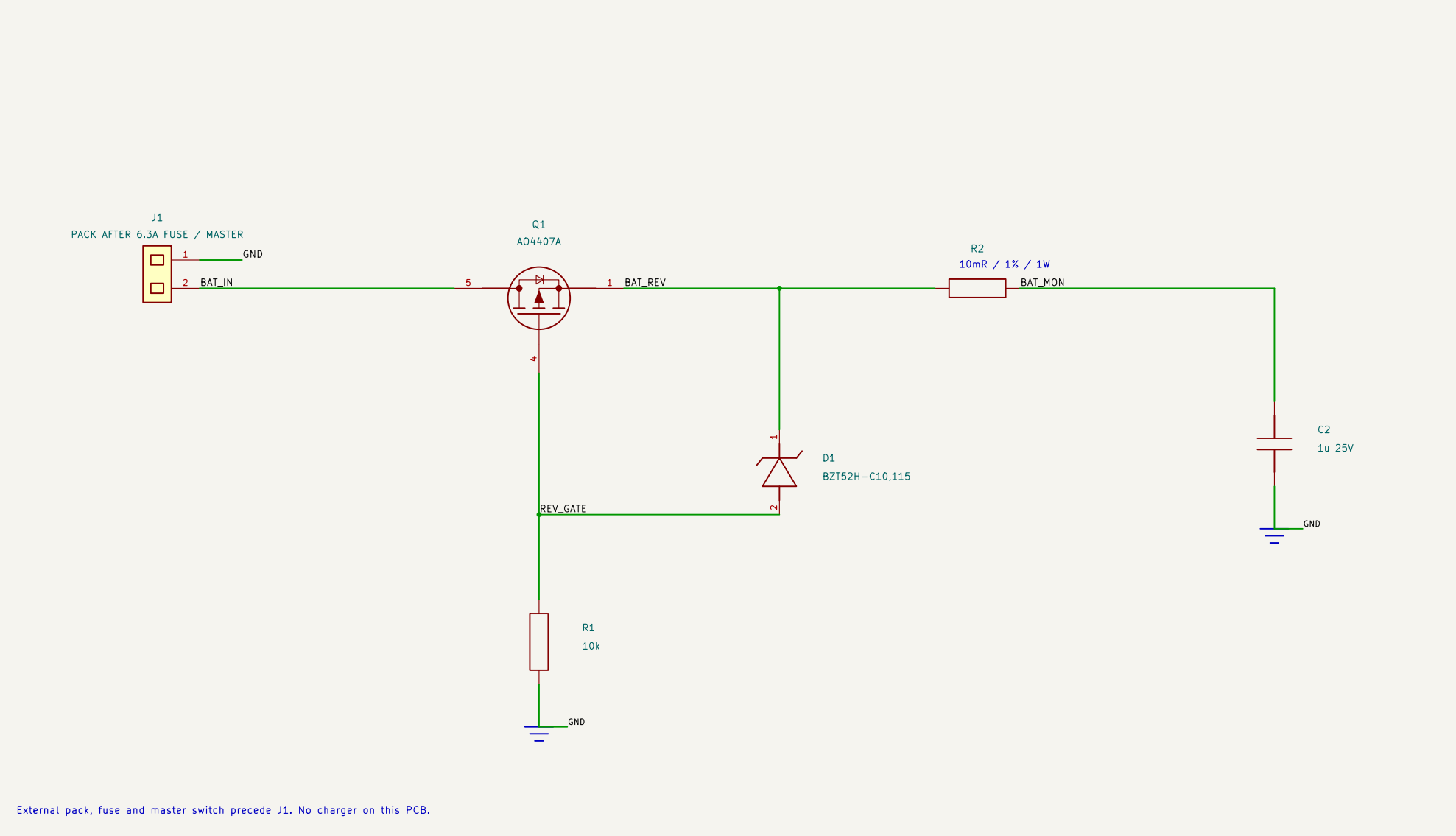


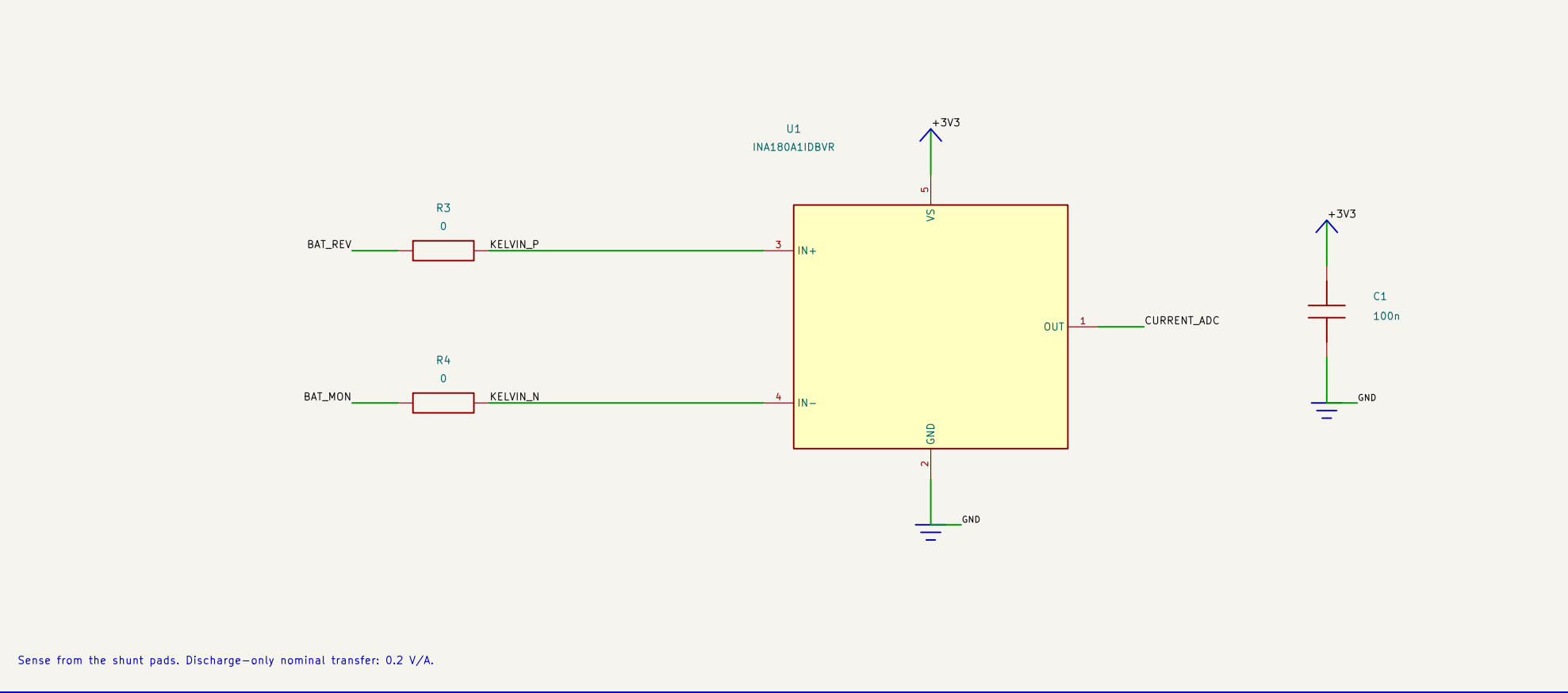
Power flows left to right. Supplies above, returns below. PROTOTYPE / UNVALIDATED.

01 Pack input / reverse protection / shunt



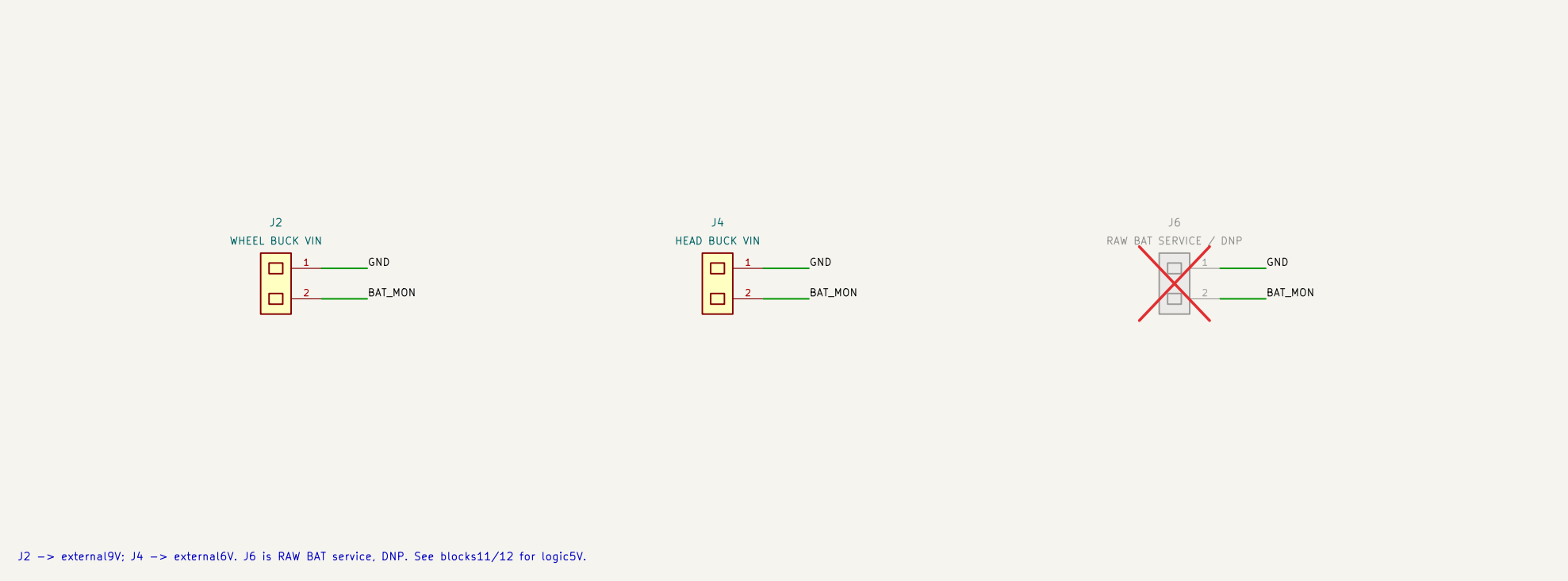
External pack, fuse and master switch precede J2. No charger on this PCB.

02 Kelvin current sense



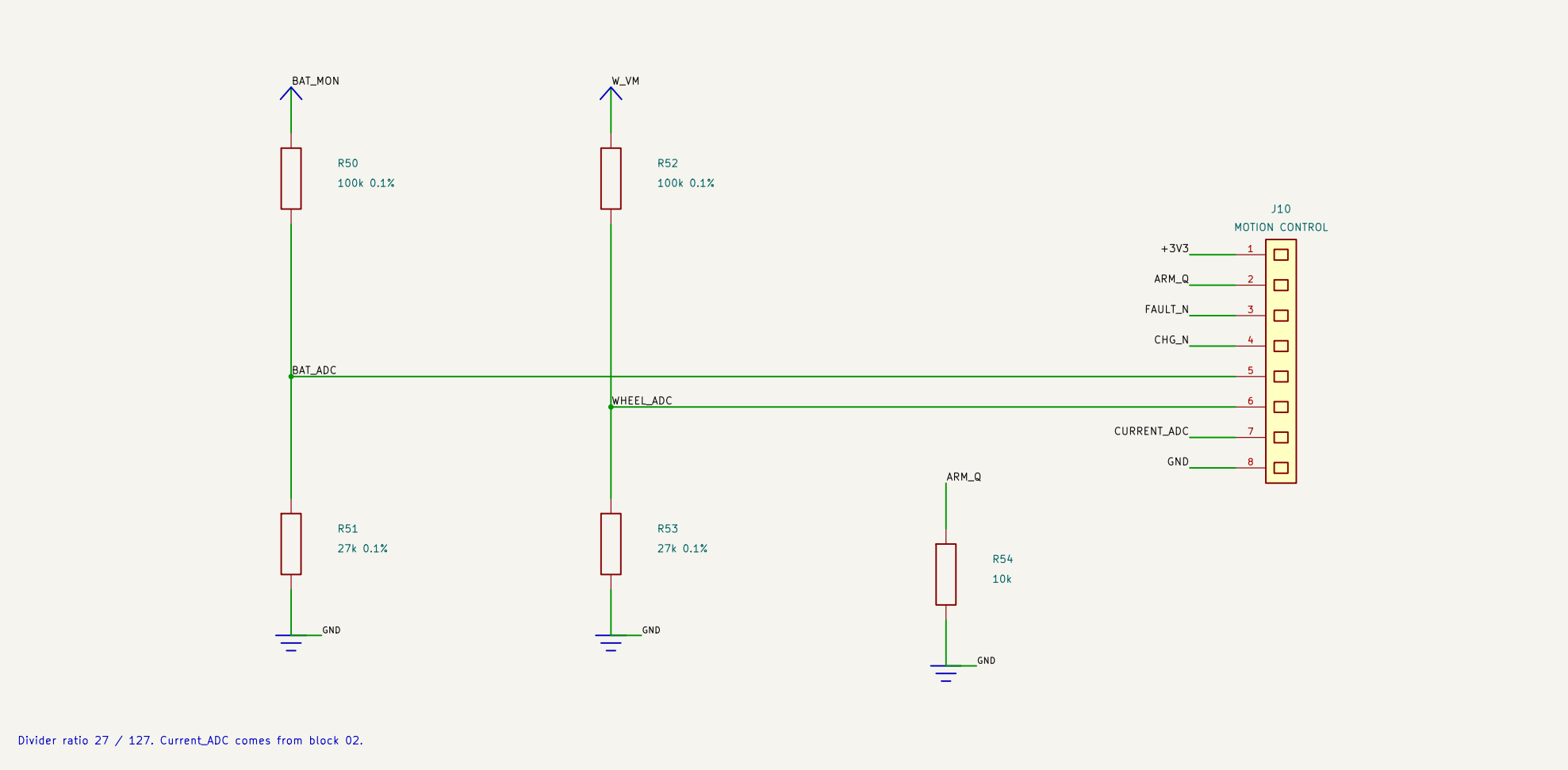
Sense from the shunt pads. Discharge-only optimal transfer: 0.2 V/A.

05 External converter input distribution



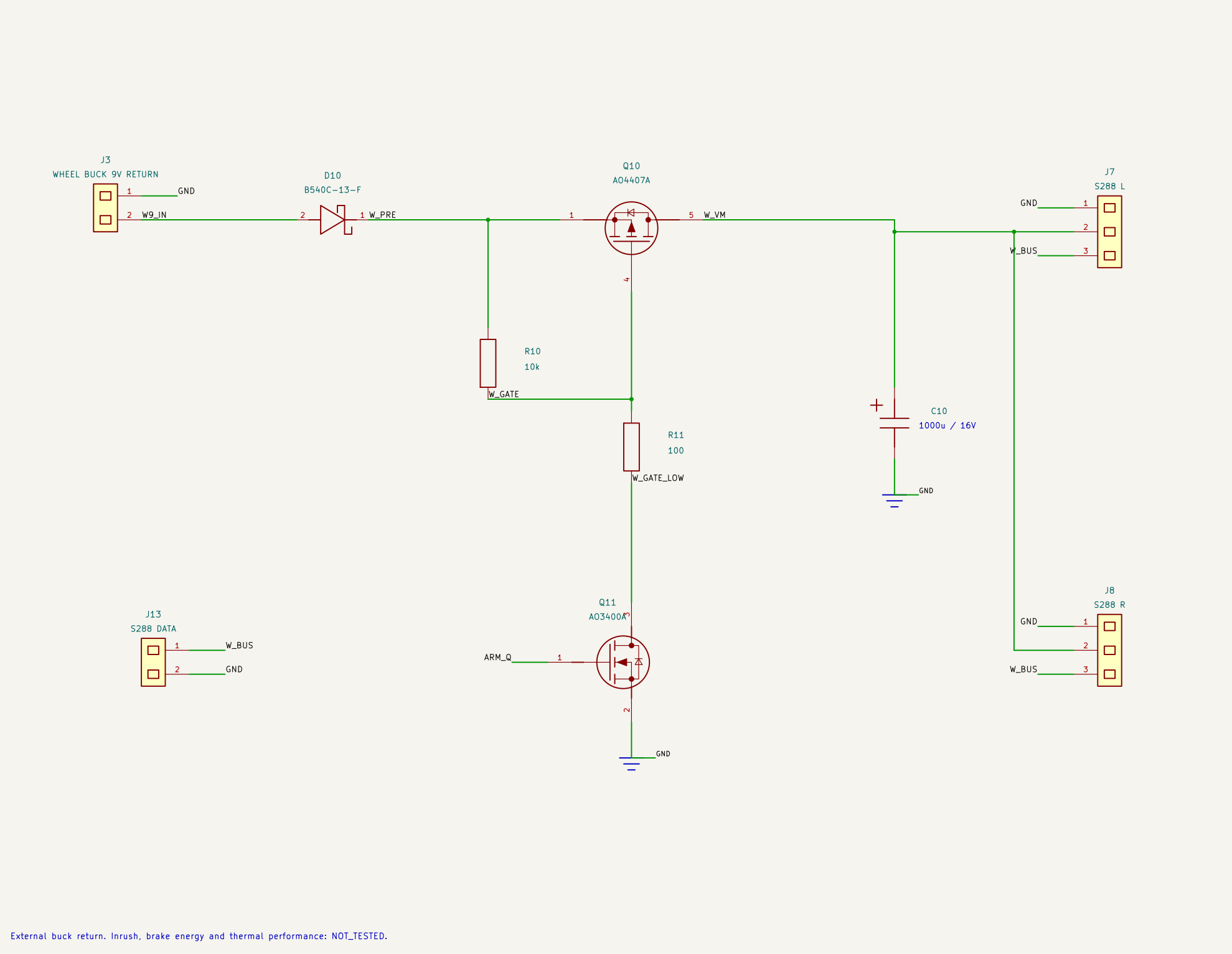
J2 -> externalVIN, J4 -> externalVIN, J6 to RAW BAT service, DNP. See blocks11/12 for logicSV.

08 Voltage monitor / motion connector



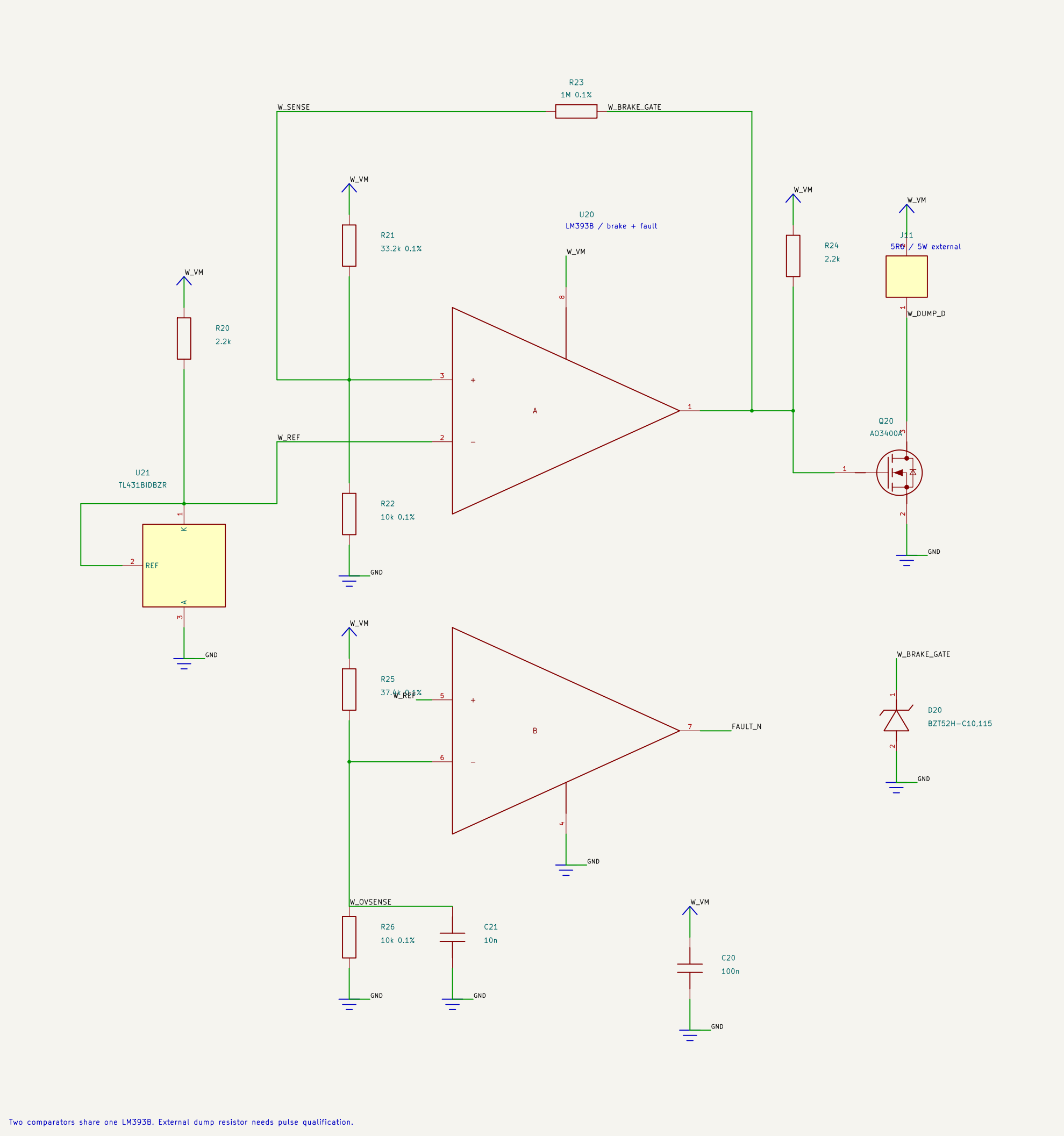
Divider ratio 27 / 127. CurrentLADC comes from block 02.

03 Wheel 9 V conditioning



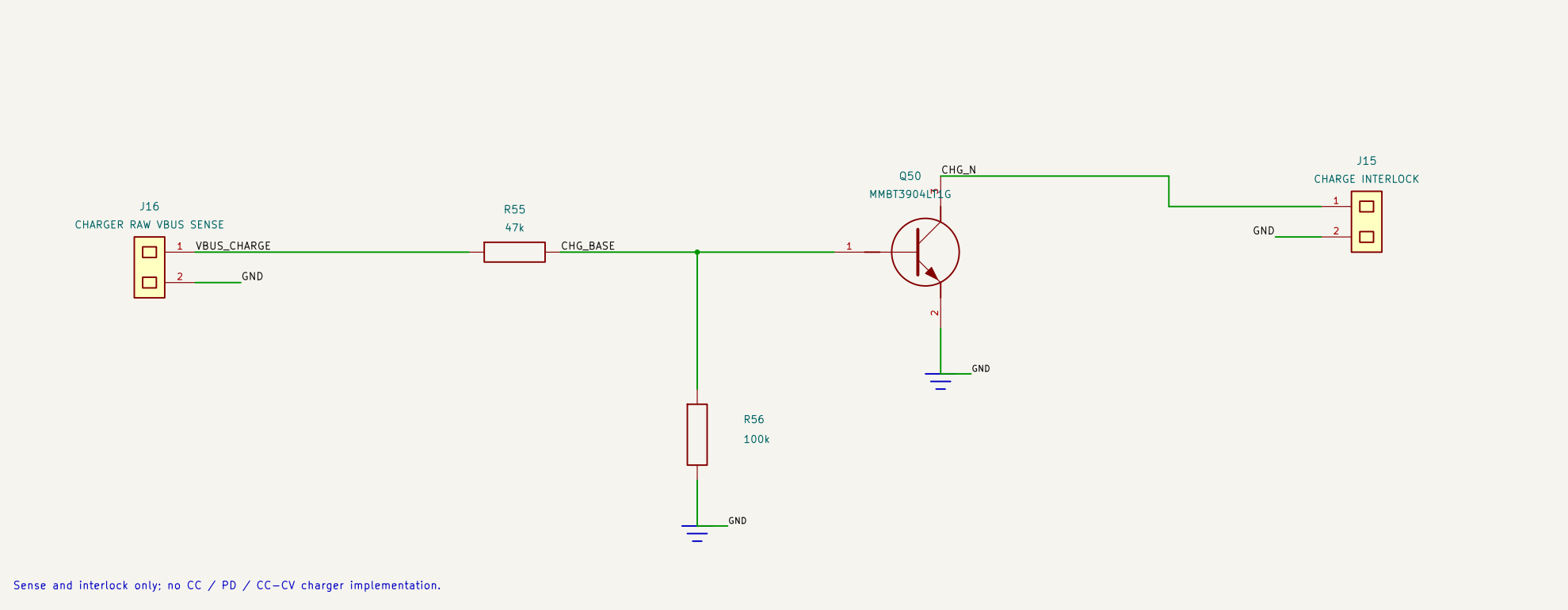
External back return, brush, brake energy and thermal performance: NOT TESTED.

06 Wheel rail / brake and overvoltage



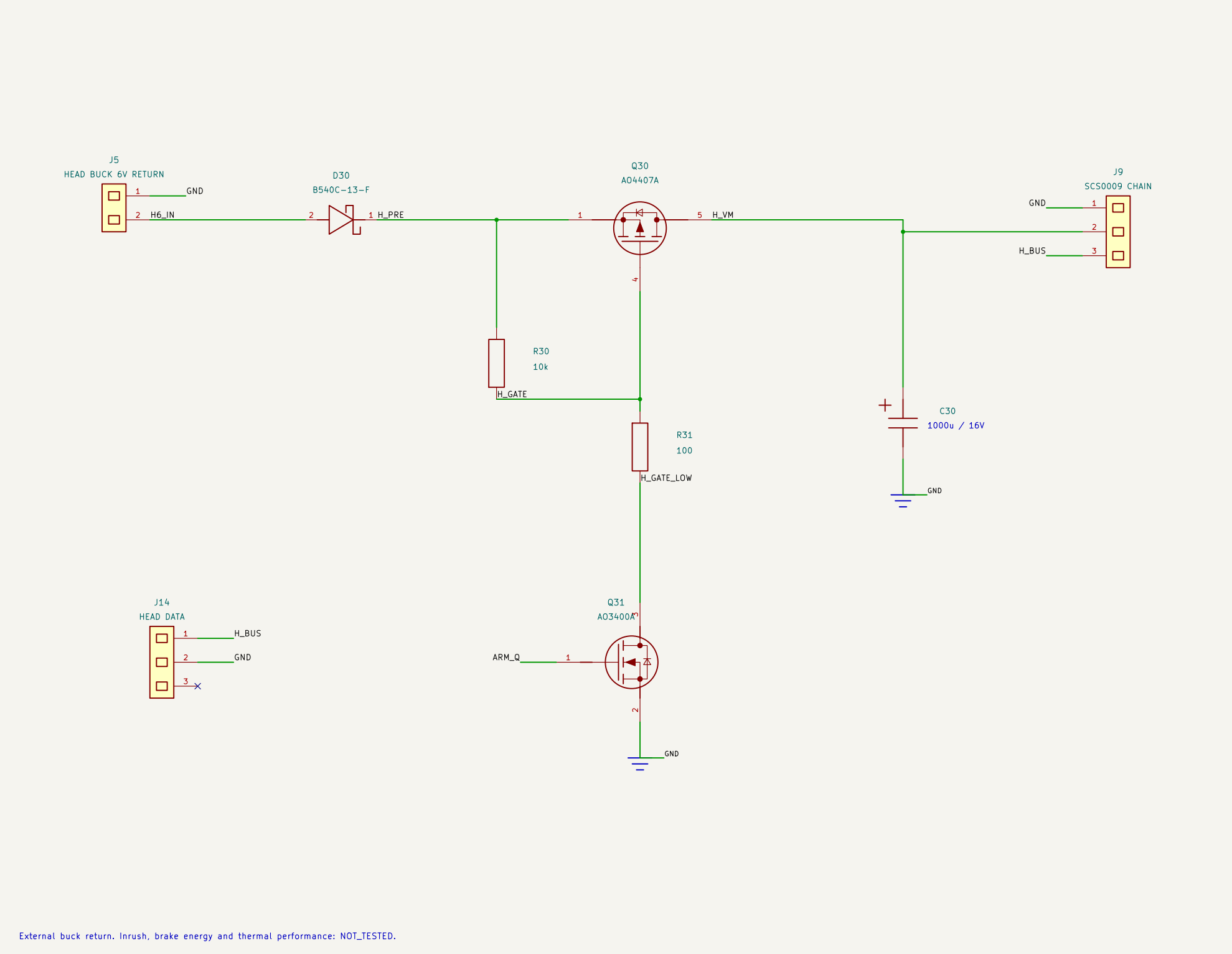
Two comparators share one LM339. External dump resistor needs pulse qualification.

09 Charge insertion / VBUS sense



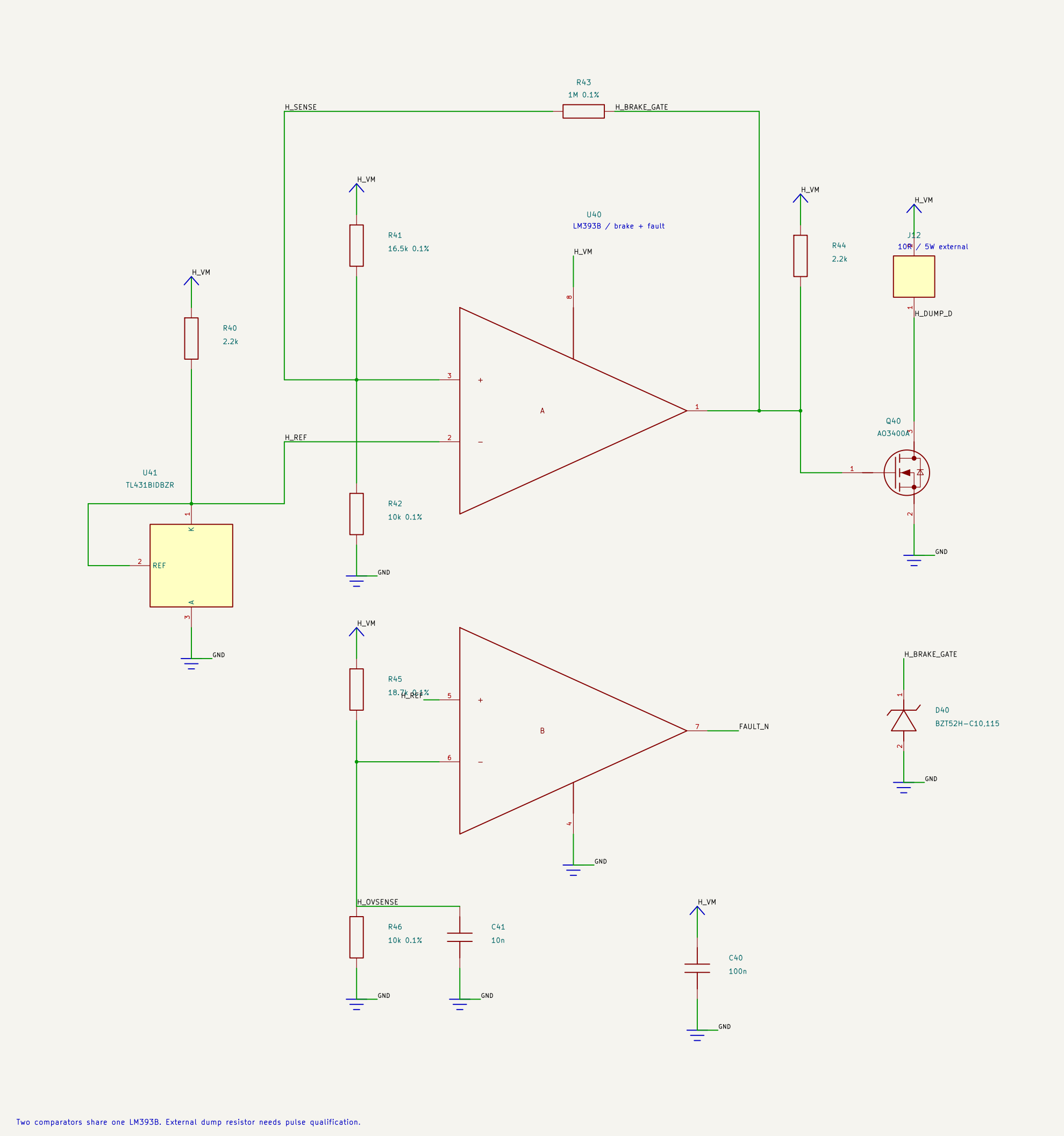
Sense and interlock only, no CC / PG / CC-VV charger implementation.

04 Head 6 V conditioning



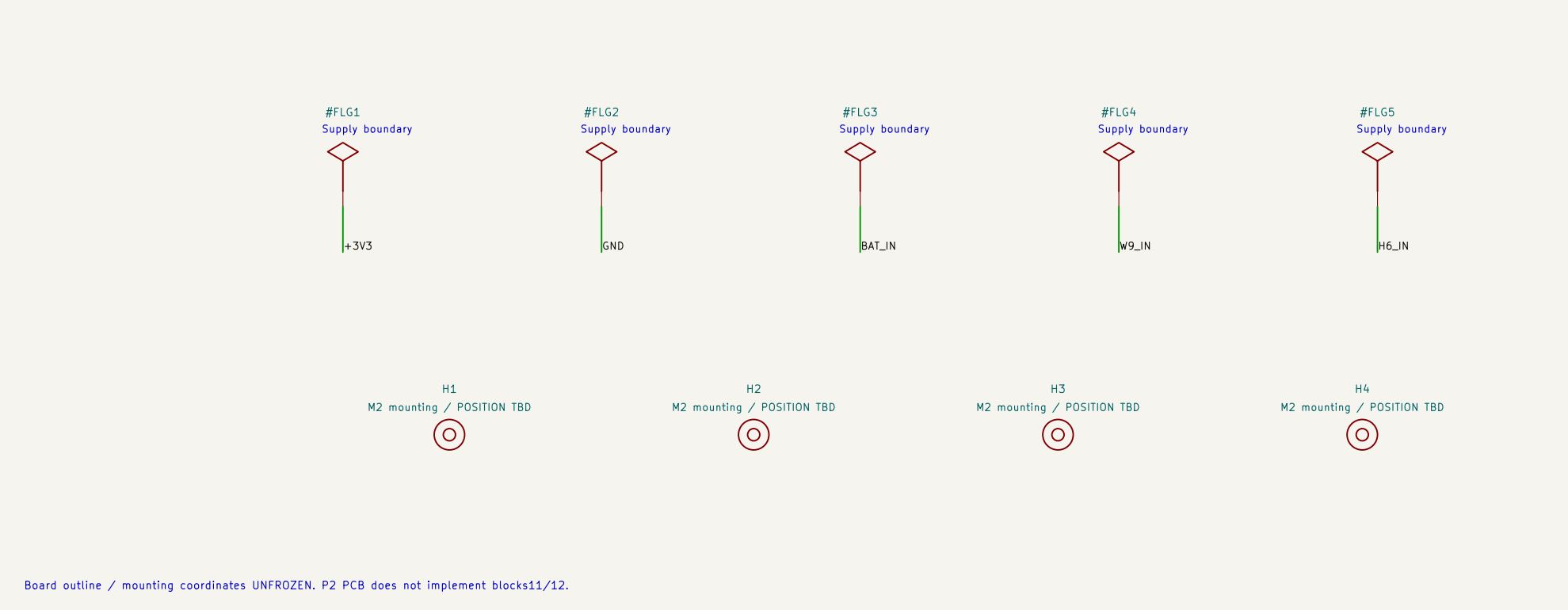
External back return, brush, brake energy and thermal performance: NOT TESTED.

07 Head rail / brake and overvoltage



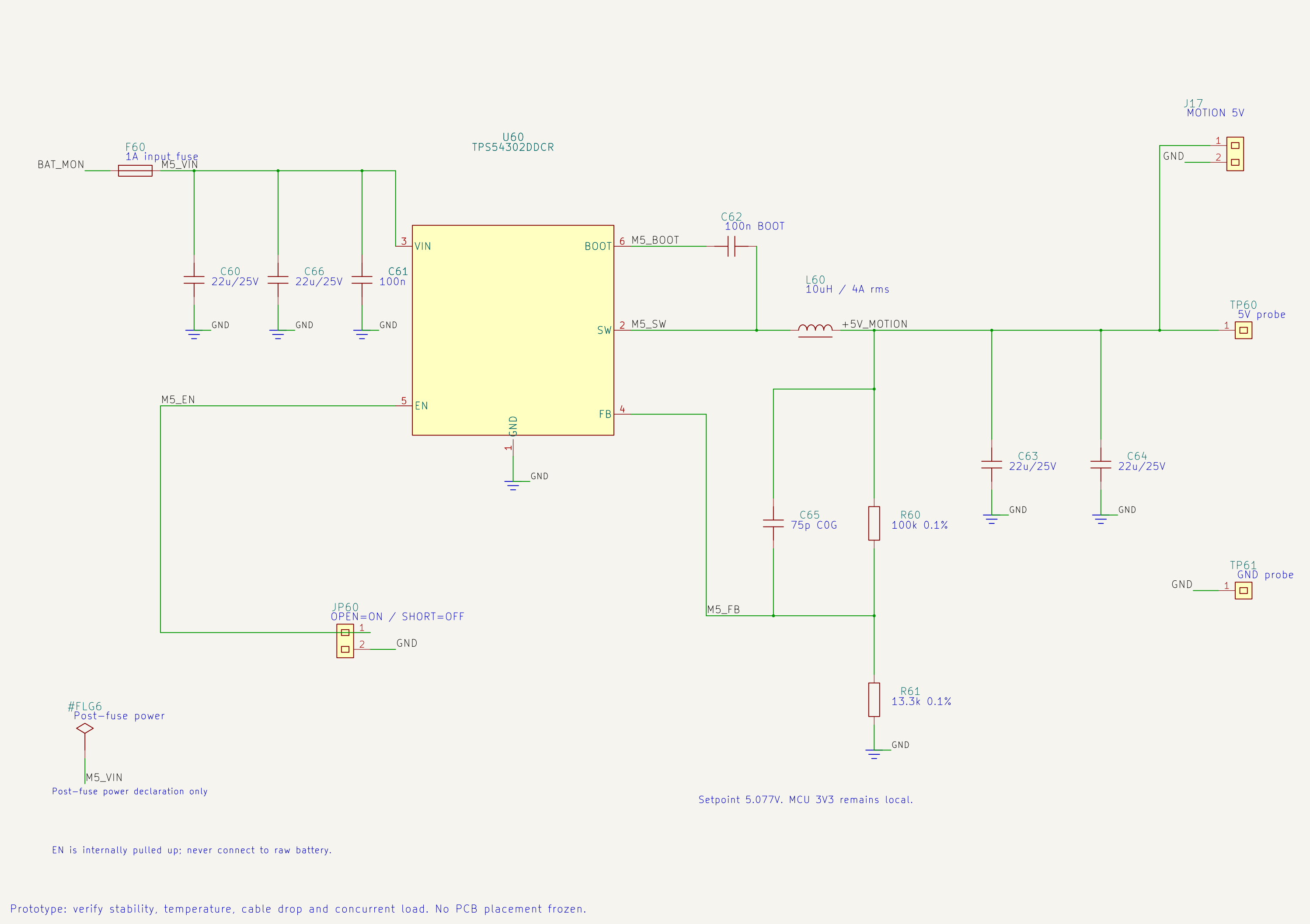
Two comparators share one LM339. External dump resistor needs pulse qualification.

10 Supply boundaries / board mounting



Board outline / mounting coordinates UNPROVEN, P2 PCB does not implement blocks11/12.

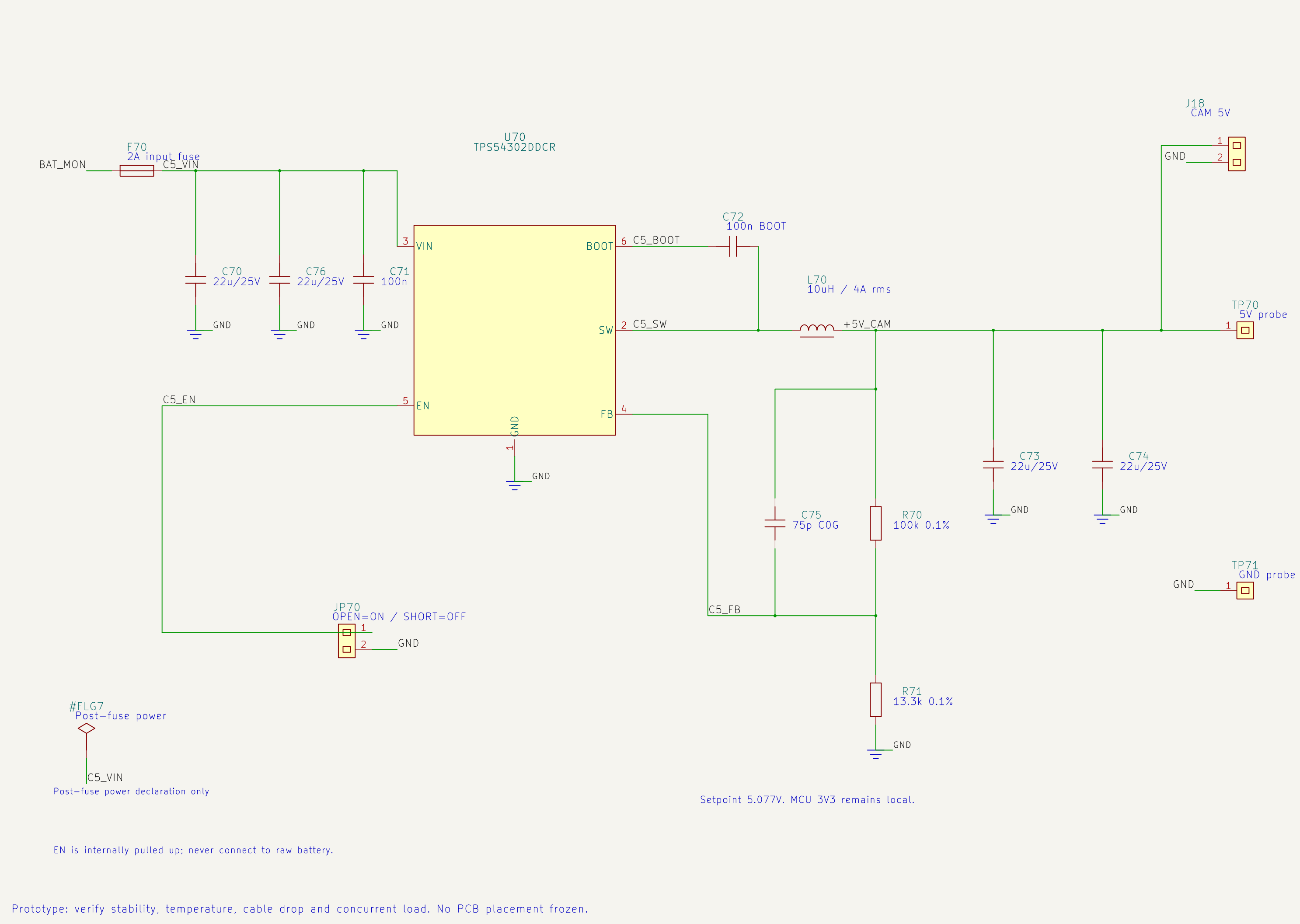
11 Motion 5 V / independent fused buck



EN is internally pulled up; never connect to raw battery.

Prototype: verify stability, temperature, cable drop and concurrent load. No PCB placement frozen.

12 Interaction 5 V / independent fused buck



EN is internally pulled up; never connect to raw battery.

Prototype: verify stability, temperature, cable drop and concurrent load. No PCB placement frozen.